

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	18 June 2026
Name	Dhanish Ladwani
Project Name	ShopEZ - E-Commerce Application
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form (username, email, password) JWT token issued on successful registration
FR-2	User Authentication	User login via email & password JWT-based session persistence across page refreshes Protected routes accessible only to authenticated users
FR-3	Product Catalog	View all products with image, title, price, and discount Filter products by gender (men / women / unisex / kids) Filter products by category Real-time keyword search across product titles Sort products by price (low-to-high / high-to-low) and by discount
FR-4	Shopping Cart	Add product with selected size to cart View cart with all items, quantities, and running total Update item quantity in cart Remove individual items from cart Clear entire cart
FR-5	Order Management	Place an order with delivery address, mobile, pincode, and payment method (COD / online) View personal order history with status and delivery date Order status: order placed → processing → shipped → delivered / cancelled
FR-6	User Profile	View profile details (username, email) Update profile information
FR-7	Admin – Dashboard	View monthly revenue analytics chart View order distribution chart View user registration statistics View best-selling products
FR-8	Admin – Product Management	Create new product (title, description, images, sizes, category, gender, price, discount) Edit existing product details Delete product from catalog
FR-9	Admin – Order Management	View all customer orders Update order status (order placed / processing / shipped / delivered / cancelled)
FR-10	Admin – User Management	View list of all registered users Delete a user account

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application must provide an intuitive and responsive UI built with vanilla CSS. All pages including the product catalog, cart, checkout, and admin dashboard must be clearly navigable. The admin

NFR No.	Non-Functional Requirement	Description
		product form must provide a live preview of the product being created.
NFR-2	Security	All API endpoints that involve user or admin data must be protected using JWT-based authentication middleware. Passwords must be hashed using bcryptjs before storage. HTTP headers must be secured via the Helmet middleware. Authentication routes must be rate-limited to 30 requests per 15 minutes; general API routes to 300 requests per 15 minutes to prevent brute-force attacks.
NFR-3	Reliability	The system must handle invalid inputs and failed API calls gracefully, returning appropriate HTTP status codes and error messages. The Express server must not crash on unhandled errors; errors must be caught and responded to consistently across all controllers.
NFR-4	Performance	API responses must be compressed using the compression middleware to reduce payload size and improve load times. The React frontend, built with Vite, must support Hot Module Replacement (HMR) in development for fast iteration. The product catalog must render search and filter results without full page reloads.
NFR-5	Availability	The backend server must be hosted such that the REST API is continuously accessible. The MongoDB database must be hosted on MongoDB Atlas with at least the M0 shared cluster tier, ensuring high availability and automated backups.
NFR-6	Scalability	The application follows a decoupled Client-Server architecture. The Express REST API is stateless (JWT-based), allowing horizontal scaling. MongoDB Atlas supports scaling to higher tiers as data volume grows. The frontend SPA can be deployed independently on any static hosting service.
NFR-7	Maintainability	The backend follows the MVC (Model-View-Controller) pattern with clearly separated routes, controllers, models, and middleware. The frontend is organised into pages, components, context, and utilities. Environment variables are managed via a .env file, keeping configuration separate from code.
NFR-8	Compatibility	The frontend must be fully compatible with the latest stable versions of Google Chrome (v90+), Mozilla Firefox (v90+), Apple Safari (v14+), and Microsoft Edge (v90+). The application must be responsive and usable on both desktop and tablet screen sizes.